Ansible Tasks - 04

- Create ansible playbook to create an ansible vault.
    - We cannot create an Ansible Vault using a playbook.
    - Because vault is about encrypting files before execution.
    - Playbooks run after everything is already loaded.

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat secret.yml
- hosts: all
  become: yes

  vars_files:
    - avengers.yml

  tasks:
    - name: Use the secret
      debug:
        msg: "{{ secret_msg }}"
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ cat avengers.yml
$ANSIBLE_VAULT;1.1;AES256
31383532653361333664323564643164646538613839383530396561306163656238623337313432
62336434316265336636353138663830306133623266343 0a313265666136313366383135313662
65643431626161346430366163356232653032636331363863613338353532313535613736666536
39386264336330656 20a6664653436326165663162343461313036633033376539623861313538 34
37653536626234303137313633303664386264333436396634626235636533653862
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook secret.yml -l web1,web2 --ask-vault-pass
Vault password:

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
adok: [web2]
ok: [web1]
TASK [Use the secret] **********************************************************************************
ok: [web1] => {
    "msg": "I am inevitable"
}
ok: [web2] => {
    "msg": "I am inevitable"
}
PLAY RECAP *********************************************************************************************
web1                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

- Write an ansible playbook to install Apache in Linux and ubuntu machine by using when condition.

```
[ec2-user@ip-172-31-24-112 ~]$ ansible all -m setup -a "filter=ansible_os_family" -l web1,web2
web2 | SUCCESS => {
    "ansible_facts": {
        "ansible_os_family": "RedHat",
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false
}
web1 | SUCCESS => {
    "ansible_facts": {
        "ansible_os_family": "RedHat",
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false
}
```

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```
[ec2-user@ip-172-31-24-112 ~]$ cat ~/playbooks/os.yml
---
- hosts: all
  become: yes

  tasks:

    - name: Install Apache on Amazon Linux
      dnf:
        name: httpd
        state: present
      when: ansible_os_family == "RedHat"

    - name: Install Apache on Ubuntu
      apt:
        name: apache2
        state: present
        update_cache: yes
      when: ansible_os_family == "Debian"
```

```
[ec2-user@ip-172-31-24-112 playbooks]$ ansible-playbook os.yml -l web1,web2

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [web2]
ok: [web1]

TASK [Install Apache on Amazon Linux] ******************************************************************************
ok: [web2]
ok: [web1]

TASK [Install Apache on Ubuntu] ************************************************************************************
skipping: [web1]
skipping: [web2]

PLAY RECAP *********************************************************************************************************
web1                       : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
web2                       : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

- Create an ansible playbook using roles to configure LAMP stack.
- LAMP stack is a set of 4 tools used to run websites:
    - L → Linux (operating system)
    - A → Apache (web server)
    - M → MySQL / MariaDB (database)
    - P → PHP (backend language)
-

```
[ec2-user@ip-172-31-24-112 project]$ ls
main.yml  roles
[ec2-user@ip-172-31-24-112 project]$ cat main.yml
- hosts: all
  become: yes

  roles:
    - apache
    - mysql
    - php
```

-

```
[ec2-user@ip-172-31-24-112 roles]$ ls
apache  mysql  php
[ec2-user@ip-172-31-24-112 roles]$
```

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```
[ec2-user@ip-172-31-24-112 project]$ ansible-playbook main.yml -l web1,web2

PLAY [all] *********************************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [web2]
ok: [web1]

TASK [apache : Install Apache] *************************************************************************
ok: [web2]
ok: [web1]

TASK [apache : Start Apache] ***************************************************************************
ok: [web2]
ok: [web1]

TASK [mysql : Install MariaDB] *************************************************************************
changed: [web2]
changed: [web1]

TASK [mysql : Start MariaDB] ***************************************************************************
changed: [web2]
changed: [web1]

TASK [php : Install PHP] *******************************************************************************
changed: [web2]
changed: [web1]

PLAY RECAP *********************************************************************************************
web1                       : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
web2                       : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

-

```
[ec2-user@ip-172-31-31-155 ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled;
     Active: active (running) since Tue 2026-03-17 14:34:38 UTC; 2min 8
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 11734 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 1067)
     Memory: 66.2M
        CPU: 474ms
     CGroup: /system.slice/mariadb.service
```

-

```
[ec2-user@ip-172-31-29-27 ~]$ systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; en
     Active: active (running) since Tue 2026-03-17 14:34:38 UTC;
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 11713 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 8 (limit: 1067)
     Memory: 66.2M
```

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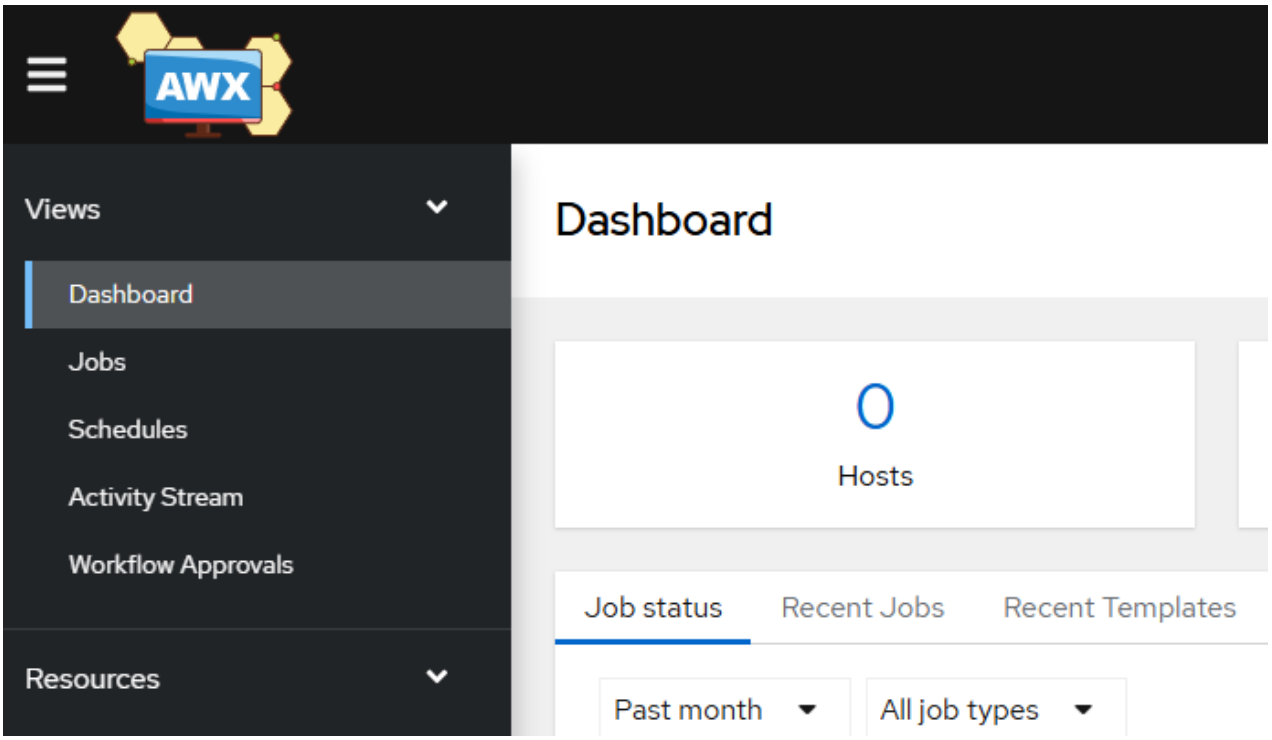
- ```
  [ec2-user@ip-172-31-24-112 roles]$ ansible all -m command -a "php -v" -l web1,web2
  web2 | CHANGED | rc=0 >>
  PHP 8.5.3 (cli) (built: Feb 10 2026 18:25:51) (NTS gcc x86_64)
  Copyright (c) The PHP Group
  Built by Amazon Linux
  Zend Engine v4.5.3, Copyright (c) Zend Technologies
      with Zend OPcache v8.5.3, Copyright (c), by Zend Technologies
  web1 | CHANGED | rc=0 >>
  PHP 8.5.3 (cli) (built: Feb 10 2026 18:25:51) (NTS gcc x86_64)
  Copyright (c) The PHP Group
  Built by Amazon Linux
  Zend Engine v4.5.3, Copyright (c) Zend Technologies
      with Zend OPcache v8.5.3, Copyright (c), by Zend Technologies
  ```
- Setup an ansible AWX and explore the options.
- ```
  [ec2-user@ip-172-31-32-57 installer]$ docker ps
  CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS                                          NAMES
  77999830df87   ansible/awx:17.1.0   "/usr/bin/tini -- /b…"   17 minutes ago   Up 17 minutes   0.0.0.0:80→8052/tcp, :::80→8052/tcp   awx_web
  da464712b95b   redis                "docker-entrypoint.s…"   17 minutes ago   Up 17 minutes   6379/tcp                                       awx_redis
  bc4c0a895276   postgres:12          "docker-entrypoint.s…"   17 minutes ago   Up 17 minutes   5432/tcp                                       awx_postgres
  [ec2-user@ip-172-31-32-57 installer]$
  ```
-